Sahaj Bhadja

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Summary

Software engineer experienced in **Python, Java, TypeScript, SQL, Linux, AWS, Docker, REST APIs, automated testing, authentication, and AI-powered products**, delivering secure, reliable cloud services in agile teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built FastAPI services and AI workflows processing 5,000+ monthly jobs with explicit state transitions, failure isolation, and validation, increasing concurrent throughput 3x while achieving 99%+ production reliability.
- Designed multi-tenant SQL storage with row-level security and indexed access patterns for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization and access-pattern tuning.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Engineered an AI grading platform with multimodal models, Python, Flask, AWS RDS, and REST APIs, serving 150+ instructors and reliably processing 10,000+ daily requests with stable API performance at scale.
- Implemented multithreaded secure ingestion for 1,000+ PDF, DOCX, ZIP, text, and image files per batch, adding automated tests, fault-tolerant handling, and stable API performance for production-scale workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed Python and PyTorch components processing 100,000+ trajectory records, improving long-horizon planning 18% through model prototyping, simulation, debugging, reproducible testing, and iterative feature delivery for production-oriented research workflows.
- Modeled real-time interactions across ego, road, and nearby-agent signals, increasing closed-loop stability 15% while evaluating edge cases and clearly documenting technical findings for senior researchers at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed cross-platform Java and Python microservices with REST APIs, AWS Glue, S3, SQL, and Docker, improving low-bandwidth response times 30% and reducing payload sizes 25% for enterprise users.
- Automated builds and 20+ unit and integration tests across Windows, macOS, and Linux using Jenkins, PyTest, and Mockito, increasing nightly data-pipeline efficiency by 68% at production scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *January 2023 – August 2023*

- Built eight Node.js microservices for authentication, permissions, notifications, resource allocation, and billing, synchronizing two million records through REST APIs, WebSockets, and Azure SQL for six core business workflows.
- Implemented JWT authentication, role-based access, encryption, firewall controls, automated testing, and Docker-based staged deployments, reducing release time 50% while maintaining secure, modular enterprise workflows across development and production environments.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, Shell, Bash
Backend: FastAPI, Flask, Node.js, Spring Boot, REST APIs, HTTP, WebSockets, microservices
Cloud/Data: AWS, Azure, Docker, Kubernetes, PostgreSQL, MySQL, Azure SQL, MongoDB, Redis
Security: JWT, authentication, authorization, role-based access control, encryption, firewalls
Quality: Git, CI/CD, Jenkins, PyTest, Mockito, Jest, unit testing, integration testing, Agile, Linux

Projects

CaptionCraft - Image Caption Generator Live | *Python, Computer Vision, AI*

- Benchmarked BLIP, GIT, and ViT-GPT-2 transformer architectures across 10,000 images and 100,000 captions, improving accuracy from 81% to 96% while building scalable, real-time inference APIs for visual understanding.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Developed Java and TypeScript services supporting 10+ databases, configuring Docker workflows across Windows, macOS, and Linux while implementing natural-language query generation, schema synchronization, and structured data access.

Big-Brain Live | *AI SaaS, Secure Cloud Platform*

- Built AI SaaS for 200+ users over 50,000+ documents using REST APIs, Clerk authentication, access controls, and real-time Convex collaboration, reducing document review time by 60% at scale.